Aim: Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

a) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "Enter the marks of English:"
read m1

echo "Enter the marks of Maths:"
read m2

echo "Enter the marks of Science:"
read m3

total=$((m1 + m2 + m3))
per=$((total / 3))

echo "Student: Total Marks = $total"
echo "Percentage = $per%"

if [ $per -gt 75 ]
then
    echo "Class: Distinction"
elif [ $per -ge 30 ]
then
    echo "Class: First Class"
else
    echo "Class: Fail"
fi
Enter the marks of English:
70
Enter the marks of Maths:
90
Enter the marks of Science:
80
Student: Total Marks = 240
Percentage = 80%
Class: Distinction
```

b) Write a menu driven shell script which will print the following menuand execute the given task.

- Display calendar of current month.

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
1
Calendar of current month:
January 2026
```

- Display today’s date and time

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
2
Today's date and time:
Sat Jan 24 22:09:38 IST 2026
```

- Display usernames those are currently logged in the system.

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
3
Users currently logged in:
HP
```

- Display your terminal number

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
4
Your terminal number:
xterm
```

C) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "Enter the value of n:"
read n

a=1
b=1

echo "Fibonacci series:"
for (( i=1; i<=n; i++ )); do
    echo -n "$a"
    if [ $i -lt $n ]; then
        echo -n ", "
    fi
    c=$((a + b))
    a=$b
    b=$c
done

echo
Enter the value of n:
6
Fibonacci series:
1, 1, 2, 3, 5, 8
```

d) Write a shell script which will accept a number b and display first n prime numbers as output

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "Enter the value of n:"
read n

count=0
num=2

echo "First $n prime numbers are:"

while [ $count -lt $n ]; do
    isprime=1

    for (( i=2; i<=num/2; i++ )); do
        if [ $((num % i)) -eq 0 ]; then
            isprime=0
            break
        fi
    done

    if [ $isprime -eq 1 ]; then
        echo -n "$num "
        count=$((count + 1))
    fi

    num=$((num + 1))
done

echo
Enter the value of n:
5
First 5 prime numbers are:
2 3 5 7 11
```

6) Write menu driven program for file handling activity

- Creation of file.

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
1
Enter file name:
Harsh
File created successfully
```

- Write content in the file.

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
2
Enter file name:
Harsh
Enter content to write (Press Ctrl+D to save):

Content written successfully
```

- Upend file content

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
3
Enter file name:
Harsh
Enter content to append (Press Ctrl+D to save):
Content appended successfully
```

- Delete file content

```
HP@DESKTOP-8HAHCL7 MINGW64 /c/OS_CD24017 (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
4
Enter file name:
Harsh
File content deleted successfully
```